

Demonstration of FabHydro: 3D Printing Techniques for Interactive Hydraulic Devices with an Affordable SLA 3D Printer

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Figure 1: Applications of FabHydro: a) Robotic Gripper; b) Fluffy Toy; c) Folding Lamp; d) Phone Stand; and e) Collaborative Game.

ABSTRACT

In this demonstration, we showcase FabHydro [6], a set of rapid and low-cost techniques to prototype interactive hydraulic devices using off-the-shelf SLA 3D printers and flexible photosensitive resin. We introduce two printing processes to seal the transmission fluid: the *Submerged Printing* process seals liquid resin inside the printed objects without assembly, and the *Printing with Plug* method which allows using variety of transmission fluid with no modification to the printer. We showcase how a range of relevant 3D printable primitives, including hydraulic generator, transmitter, and actuator can be printed and combined to create interactive examples.

CCS CONCEPTS

• **Human-centered computing** → *Graphical user interfaces; User interface design; User centered design*; • **Software and its engineering** → *Software design techniques*.

KEYWORDS

Fabrication, 3D Printing, Interaction, Design

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1 INTRODUCTION

One of the long-term visions for additive manufacturing is to print devices with functionalities and interactivities [5]. For example, Peng et al. [3] designed a custom 3D printer that can embed magnetic wires into the printing process to build reluctant motors. MacCurdy et al. [2] proposed to seal droplets inside a printed cavity to make hydraulic walking robots. These approaches show the potential to print one-off objects with mechanical motion, but the fabrication process are often challenging. They either require the end users to build custom 3D printers with complex hardware designs, or require high-end industrial 3D printers that may cost over 200,000 US dollars and thus not accessible by many.

In this demonstration, we showcase FabHydro [6], a set of fabrication methods to democratize the printing of hydraulic-driven mechanical devices with consumer-grade 3D printers that cost less than 200 US dollars. Specifically, We showcase two printing methods to print actuatable hydraulic systems. With the first method, named *Submerged Printing*, we demo how to seal liquid resin inside a 3D printed structure without post-assembly. With the second method, *Printing with Plug*, we demo how to include variety of transmission fluid into the printed devices without modification to the printer. We conclude with a set of examples such as a robotic gripper and a folding lamp that can be directly printed with the FabHydro methods.

2 FABHYDRO OVERVIEW

FabHydro focuses on 3D printed hydraulic systems, including the generator, transmitter, and actuator, that can be fabricated with single material using affordable SLA 3D printers. The system is printed with an Elegoo Mars LCD-SLA 3D Printer [1] with F39/F69 flexible photosensitive resin from RESIONE [4].

The successful print of a hydraulic device needs to consider two aspects: the print of deformable structures (e.g. bellows structure) and how to seal the liquid inside.

With the default factory setup and printing parameters, deformable hydraulic structures cannot be successfully printed with the off-the-shelf SLA 3D printer. Print failures are partially caused

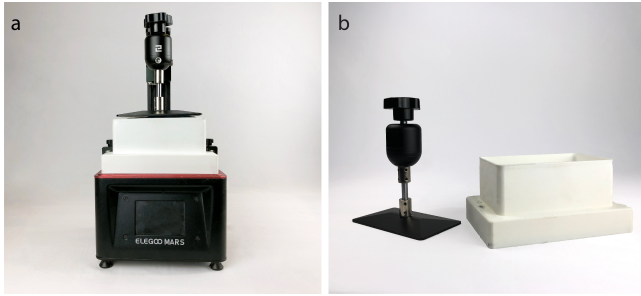


Figure 2: Printer Modification: a) modified printer; and b) extended printing plate and modified vat.

by undesired elongation of the printed structures and flexible overhangs due to the flexible resin material. To solve these problems, we proposed to use customized support structures and also custom printer setting with longer exposure times per printed layer. Please refer to our full paper for more details.

To seal hydraulic fluid inside the printed structures, we propose two approaches *Submerged Printing Process* and *Printing with Plug*. We now discuss each of the process below.

Submerged Printing Process. With the submerged printing process, we print a complete hydraulic device with sealed liquid inside that requires minimal post process. This requires lightweight modifications to the current 3D printer hardware. Our solution is to make the entire printing process as a whole happens below the liquid level (Figure 2 a). The model is still ‘grow’ upside-down from the building plate but will not be pulled out of the liquid until the print is finished.

Specifically, we propose a low-cost modification with a new vat design with extended walls. Since this will block the factory building plate’s dipping motion, we further extended the building plate with a pair of rod couplers (Figure 2b left).

Printing with Plug. This approach requires injecting hydraulic fluid after the printing but does not require hardware modification. We include threaded intake holes and companion plugs to the hydraulic structure design. After printing, users can fill the printed structure with hydraulic fluid such as water, oil, or other liquid from the threaded hole. The companion plug can be used to block the intake. Note that since this method does not aim for no-assembly printing, the designer can print several components respectively and assembly them afterward. This can be useful if the hydraulic design goes beyond the volume of the printer.

3 APPLICATIONS

To illustrate the potential of our approach and highlight our design space, we showcase five applications, each emphasizing one or more features of FabHydro. For more detailed on the FabHydro design opportunity, please refer to [6].

Robotic Gripper with Distant Control. We demonstrate the fabrication of a robotic gripper (Figure 1 a) that can be controlled at a distance with hydraulic energy transmission. With the long transmission distance, the grippers still maintain the strength of

carrying a cake. It can also help handle hazardous material with manual control from distance.

Movable Fluffy Elephant. To demonstrate the use case of a printed switch, we present a soft fluffy toy (Figure 1 b) with binary switch to vary actuation in two different end result. The user can choose to actuate all three actuators at once or actuate them individually by turning the switch on or off.

Folding Lamp. We also fabricated a folding lamp (Figure 1 c) with controllable posture shows the technology enables hydraulic control towards circuitry. The user can adjust the amount of unfolding by pressing the generator to different heights. LED is attached to the lamp with copper tape, which closes the circuit when actuation happens.

Phone Stand as a Tangible UI. This phone stand raises user’s phone upon alarm and phone call as a shape changing interface (Figure 1 d). Unlike previous examples, the phone stand’s actuator is designed with widened cross-section to hold phones properly. The expansion units are also not evenly distributed. Instead, the distance increases gradually, creating a bending surface with tangent continuity at the top.

Tangibles for Collaborative Game. We also formed a collaborative game (Figure 1 e) to show the interactive opportunity provided by the instantaneous of the actuation. In this example, the tangible prop is designed with a shared connection to two parallel extending actuators; each actuator is then connected to the cat paw generator. The game’s goal is for the two users to catch fish by placing the prop at the correct location collaboratively.

4 CONCLUSION

In this demo, we presented FabHydro and its applications. With FabHydro, sealed 3D structures can be printed either submerged or with plugs using flexible photosensitive resin and custom panel supports. We discussed the printing strategy and showed the breadth of our approach with a list of examples. With the low-cost setup and accessible workflow, this work will further enrich the design space and design tools available to designers for novel haptic user interfaces and interactive devices.

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